- T/W_{TO}
- W_{TO}/S

$W_{TO} \approx 2400$ lb

- Wetted Aspect Ratio when L/D max Approx. (Good for quick estimate)

$L \doteq W$, $\dfrac{W}{AR + \dfrac{AR}{S_{wet}/S_{ref}}}$

- $AR_{effective} = AR\left(1 + \dfrac{1.9h}{b}\right)$

Max efficiency: $L/D_{max} = K_{LD}\sqrt{(AR_{wet})}$

- K_{LD} = Statistical Loiter parameter
- ~~AR_{wet} = Total wetted area / wing ref. area~~
- $AR_{wet} = \dfrac{b^2}{\text{total wetted area}}$
- b = wing span
- $AR_{effective} = AR\left(1 + \dfrac{1.9h}{b}\right)$ A process of designing aircraft

"Full" drag Polar Build Up Starts HERE

- Wing sweep $\Lambda_{c/4}$
- $AR = b^2/S$, Each calculation needs to specify what type of S, wing area, the span & average chord length & change AR calculations
- Oswald efficiency factor, when $\Lambda_{LE} = 0°$, $e = 1.78(1 - 0.045(AR)^{0.68}) - 0.64$

when $\Lambda_{LE} > 30°$: $e = 4.61(1 - 0.045(AR)^{0.68})(\cos\Lambda_{LE})^{0.15} - 3.1$

When $0 < \Lambda_{LE} < 30°$: Linearly interpolate between 2 eq-n

1

- Assume $C_{D,min}$
- Assume $C_{L,min}$

$C_L = \dfrac{2W}{\rho \cdot V^2 \cdot S}$, steady flights

2

- Quadratic lift term $K_1 = 1/(\pi \cdot AR \cdot e)$
- Linear Lift term $K_2 = -2K_1 C_{L,min}$
- Zero-lift coefficient of drag $C_{D0} = C_{D,min} + K_1 C_{L,min}^2$
- $C_D = C_{D0} + K_1 C_L^2 + K_2 C_L$

Energy Based Constraint Analysis

- Phase 1: Takeoff
  - TO field length S_G
  - Safety factor (in all) K_{TO}
  - max $C_L = \dfrac{2W}{\rho \cdot V_{stall}^2 \cdot S}$

$\dfrac{T}{W} = \dfrac{\beta^2}{\alpha}\left[\dfrac{K_{TO}^2}{S_G \rho g C_{L\,max}}\right]\left(\dfrac{W_{TO}}{S}\right)$

- Phase 2: Climb
  - max altitude above field
  - climb speed
  - time-to-climb

$= \dfrac{\beta}{\alpha}\left[\dfrac{K_1\beta}{q}\left(\dfrac{W_{TO}}{S}\right) + K_2 + \dfrac{C_{D0}}{\frac{\beta}{q}}\left(\dfrac{1}{\frac{W_{TO}}{S}}\right) + \dfrac{1}{V}\dfrac{d(h)}{dt}\right]$

$= \dfrac{\beta}{\alpha}\left[\dfrac{K_1\beta}{q}\left(\dfrac{W_{TO}}{S}\right) + K_2 + \dfrac{C_{D0}}{\frac{\beta}{q}}\left(\dfrac{1}{\frac{W_{TO}}{S}}\right)\right]$

- Phase 3: Cruise
  - Altitude above field

Best range Cruise speed Calculated earlier

- Phase 4: Turn
  - Altitude above field
  - turn radius

$\dfrac{\beta}{\alpha}\left[\dfrac{K_1\beta n^2}{q}\left(\dfrac{W_{TO}}{S}\right) + K_2 n + \dfrac{C_{D0}}{\frac{\beta}{q}}\left(\dfrac{1}{\frac{W_{TO}}{S}}\right)\right]$

Best loiter Endurance speed Calc Earlier